

Program 1: Creating and Using a User-Defined Package

Step 1: Create a package class

```
// File: Calculator.java
package mypackage;

public class Calculator {
    public int add(int a, int b) {
        return a + b;
    }

    public int sub(int a, int b) {
        return a - b;
    }
}
```

Step 2: Create main class to access package

```
// File: MainApp.java
import mypackage.Calculator;

public class MainApp {
    public static void main(String[] args) {
        Calculator obj = new Calculator();

        System.out.println("Addition: " + obj.add(10, 5));
        System.out.println("Subtraction: " + obj.sub(10, 5));
    }
}
```

Compilation and Execution

```
javac -d . Calculator.java
javac MainApp.java
java MainApp
```

Output

```
Addition: 15
Subtraction: 5
```

Program 2: Using Fully Qualified Name (Without import)

```
// File: Demo.java
public class Demo {
    public static void main(String[] args) {
        mypackage.Calculator obj = new mypackage.Calculator();

        System.out.println("Addition: " + obj.add(7, 3));
    }
}
```

Output

Addition: 10

Program 3: Creating and Using a Sub-Package

Step 1: Create class in subpackage

```
// File: Message.java
package com.company.utils;

public class Message {
    public void display() {
        System.out.println("Hello from subpackage");
    }
}
```

Step 2: Access subpackage class

```
// File: Main.java
import com.company.utils.Message;

public class Main {
    public static void main(String[] args) {
        Message msg = new Message();
        msg.display();
    }
}
```

Compilation

```
javac -d . Message.java
javac Main.java
java Main
```

Output

Hello from subpackage

Program 4: Accessing Classes from Another Package

```
// File: A.java
package pack1;

public class A {
    public void show() {
        System.out.println("Inside Package pack1");
    }
}

// File: B.java
package pack2;

import pack1.A;

public class B {
    public static void main(String[] args) {
        A obj = new A();
        obj.show();
    }
}
```

Compilation

```
javac -d . A.java
javac -d . B.java
java pack2.B
```

Output

Inside Package pack1